

Fundamental of Machine Learning Project Guidelines and Rubric

National University, Bangladesh

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1 Objective

The project aims to help students apply machine learning techniques to solve a real-life problem that they personally observe in their workplace, community, or society. Students will demonstrate their ability to identify a meaningful problem, collect and analyze original data, and design a suitable ML-based solution.

2 Fundamental Machine Learning Project Guide lines

As a part of the course, you have to submit a project. Please select a project considering the following aspects:

1. Problem Selection

- Choose a real-world problem that is relevant and meaningful to you (e.g., something from your workplace, local environment, or daily life).
- The problem must be specific, measurable, and solvable using machine learning methods.
- Clearly define the problem statement and objectives.
- Why is this problem important?
- What insights do you aim to derive?

2. Data Creation

- You must collect or generate your own dataset.
- The dataset must not be downloaded from any public repository (e.g., Kaggle, UCI, GitHub, Google Datasets, etc.).
- Acceptable sources:

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- Data collected through surveys, sensors, internal databases, or manual records.
- Data you record from personal observations or local

systems. • Ensure that your dataset:

- Contains enough samples for training and testing.
- Is clean, consistent, and properly structured.
- Complies with ethical and privacy standards — avoid sensitive or personally identifiable information.

3. Data Preprocessing and Cleaning

- Handling missing values (imputation, removal).
- Handling duplicates (removing redundant data).
- Dealing with outliers (using IQR, Z-score, etc.).
- Data transformation (scaling, normalization).
- Feature engineering (creating new meaningful

features). 4. Model Selection and Implementation

- Analyze the problem and select the most appropriate machine learning technique (e.g., classification, regression, clustering, time series analysis).
- You may test multiple models and compare their performance (e.g., Decision Tree vs. Random Forest, Linear vs. Lasso Regression).
- Explain the reasoning behind model selection.

5. Evaluation

- Use suitable evaluation metrics (accuracy, precision, recall, RMSE, R^2 , etc.) depending on the problem type.

6. Results Interpretation

- Performance metrics (accuracy, precision, recall, RMSE, etc.) should be shown using tables, charts, and graphs.
- Discuss which model performed best and why.

7. Conclusion and Recommendations

- Summary of key findings.
- How can insights be applied to solve real-world problems? • Future scope for improvement.
- Any limitations of the analysis.

8. Code & Documentation

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- Well-commented and structured code (Python).
- README file explaining the project workflow.

- Jupyter Notebook or report (if applicable).

9. Report Structure

- Title Page – Project title, student name, ID, course, date.
- Abstract – Short summary of the project (150–200 words).
- Introduction and Problem Statement – Describe the problem and its importance.
- Dataset Description – Explain how you collected or generated the data.
- Data Preprocessing – Describe cleaning, feature engineering, or transformations performed.
- Model Development – Discuss the algorithms used and justify your choices.
- Results and Comparison – Present performance metrics and comparisons between models.
- Findings and Discussion – Interpret results and insights gained from the project.
- Future Work / Plan – Suggest how the project can be improved or expanded in the future.
- References (if any) – Cite any resources or tools used (excluding datasets).
- Submission Requirements:
 - Report: Well-formatted, maximum 10–12 pages (excluding appendices).
 - Code: Submitted as a '.py' or '.ipynb' file (or zipped folder).

3 Fundamental of Machine Learning Project Rubric The

rubric of the project is shown in Table ??.

Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)
Clearly defined, original, and highly relevant real world problem; strong justification for using ML.	Problem is defined and relevant but lacks strong rationale or novelty.	Problem is somewhat relevant but vague or not well motivated.	Problem unclear or not suitable for ML application.
Dataset fully created by student, clean, sufficient, and ethically collected; excellent documentation.	Dataset mostly original and relevant; minor issues with quality or documentation.	Dataset partially original or has notable quality gaps.	Dataset copied, incomplete, or poorly collected.
Appropriate algorithms chosen with clear reasoning and alignment to problem type.	Model selection reasonable with minor justification gaps.	Limited explanation of model choice; partial relevance.	Inappropriate or unjustified model selection.
Excellent data preprocessing, feature selection, and evaluation; strong analytical rigor.	Good implementation with minor gaps in analysis.	Adequate preprocessing but limited depth in analysis.	Weak or incorrect implementation; major analytical flaws.

Clear, accurate comparison between models with correct metrics and insightful discussion.	Good comparison but lacks depth or clarity in interpretation.	Some comparison but incomplete or inconsistent metrics.	No meaningful comparison or incorrect analysis.
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Criteria Score (/100)

Problem Identification & Relevance

/20

Dataset Originality & Quality

/15

Model Selection & Justification

/20

Implementation & Analysis

Results & Comparison
/10

/10

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Table 1: Rubric for Machine Learning Project Assessment (Continued)

Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)
Deep insight into results; demonstrates critical thinking and strong interpretation.	Good findings with logical discussion; minor depth issues.	Basic findings; limited discussion of implications.	Minimal or unclear interpretation of results.
Practical, innovative, and well-explained future improvements.	Reasonable future plan with some missing details.	General suggestions without clear direction.	No future plan or unrealistic suggestions.
Exceptionally clear, well organized, technically sound, and visually appealing report.	Good structure and clarity; minor formatting issues.	Acceptable report but lacks polish or consistency.	Poorly organized, unclear, or incomplete documentation.

Criteria Score (/100)

Findings & Discussion

Future Work &
Plan

/5

Report Presenta
tion & Documenta
tion

/10

Final Score: 100

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/10